

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Computer Vision with OpenCV **COURSE CODE:** DSA02

COURSE OUTCOME COVERED

CO2: Apply image processing and feature extraction techniques, including edge detection, motion estimation, and optical flow, for analyzing images.. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5 Total Marks:50 Annexure A

Analytical Problem Solving

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Numerical Problems						
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Problem Underst and ing	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	

Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	
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Annexure A

Analytical Problem Solving

Question 1: Salt-and-Pepper Noise Removal Before Edge Detection Scenario: A grayscale image of size 1024×1024 is captured using a surveillance camera under poor lighting conditions. The image is corrupted by salt-and-pepper noise with a noise density of 0.05. This means that some pixels are randomly changed to either black or white intensity values. The image has to be preprocessed before applying an edge detection algorithm. Since edge detection is sensitive to abrupt intensity variations, the selected filter should remove noise effectively while preserving important edge details.

Answer the following:

- Identify the most suitable filtering technique for removing salt-and-pepper noise without causing significant edge blurring.
- Explain why the selected filter performs better than an averaging filter for impulse noise removal.
- If a 3×3 median filter is applied, estimate the probability that a noisy center pixel will be corrected. Clearly state your assumptions.
- Estimate the total number of noisy pixels present in the original image.
- Estimate the expected number of noisy pixels remaining after median filtering.
- Explain how the preprocessing strategy would change if the same image was affected by Gaussian noise instead of salt-and-pepper noise.

Question 2: Discretization of a Continuous Image

Scenario : An analog grayscale image has a smooth sinusoidal intensity variation along the horizontal direction. The intensity changes continuously with respect to the spatial coordinate x , while it remains constant along the y -direction. The image is defined over a spatial range of 0 to 255 in both directions. This continuous image has to be converted into a digital image for storage and further processing in a computer vision system.

Answer the following:

- Explain how the continuous image can be converted into a digital image using sampling and quantization.
- Identify the spatial variation pattern of the image in the x -direction.
- Determine the minimum sampling requirement needed to avoid aliasing in the x direction.
- Explain what will happen if the image is sampled below the required sampling rate.
- Suggest a suitable 8-bit quantization scheme for storing the image.
- Discuss the effect of quantization on image quality, storage size, and intensity accuracy.

Question 3: Edge Detection in Thin Line Images

Scenario: A 512×512 grayscale image contains very thin vertical and horizontal white lines of 1-pixel thickness on a dark uniform background. Such images are commonly used in document analysis, engineering drawing interpretation, circuit diagram processing, and road-lane detection.

The objective is to detect the line boundaries accurately without losing thin edge information.

Answer the following:

- Compare Sobel, Prewitt, and Canny edge detectors for detecting 1-pixel-thick vertical and horizontal lines.
- For a horizontal edge located near pixel position (256, 256), compute the Sobel gradient response using the standard Sobel operator.
- Explain how the gradient magnitude and gradient direction help in identifying the strength and orientation of an edge.
- Discuss the role of Gaussian smoothing in the Canny edge detection process. e. Explain how Gaussian smoothing may affect edge localization, especially for thin lines.
- If the Canny threshold is reduced to half of its original value, explain the expected changes in detected edges, weak edges, and false edges.

Question 4: Feature Extraction for Handwritten Digit Recognition Scenario: A computer vision system is being developed for recognizing handwritten digits from scanned images. The handwritten digits vary in size, orientation, stroke thickness, and writing style. Therefore, robust feature extraction is required before classification.

Answer the following:

- Explain the working principles of SIFT, SURF, and ORB feature extraction methods.
- Compare SIFT, SURF, and ORB in terms of scale invariance, rotation invariance, computational cost, and real-time suitability.
- If a digit image is rotated by 45 degrees, identify which feature extraction methods can still produce reliable matching features.
- If the average number of keypoints detected per image is 120 and 50 images are processed, compute the total number of keypoints detected.
- Suggest a suitable dimensionality reduction technique to reduce the feature size while retaining 95% of the useful information.
- Explain how dimensionality reduction improves classification efficiency in handwritten digit recognition.

Question 5: Morphological Preprocessing for Character Recognition Scenario: A binary image of size 256×256 contains scanned handwritten characters. Due to poor scanning quality, the image contains small white noise dots in the background and small black gaps inside the character strokes.

Before applying character recognition, the image must be cleaned using morphological preprocessing techniques.

Answer the following:

- Identify the morphological operation suitable for removing small white noise dots

from the background.

b. Identify the morphological operation suitable for filling small gaps or holes inside the character strokes.

c. Explain the difference between opening and closing operations using simple image processing terminology.

d. If a 3×3 square structuring element is used, explain how it affects thin character strokes.

e. Discuss the risk of using a large structuring element during preprocessing. f.

Propose a complete preprocessing pipeline for handwritten character recognition, starting from image acquisition to feature extraction.